SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1706**

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

ARTICLE REVIEW

Code of Conduct:

I Rekha S K (192511023) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Rekha S K

Annexure A

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

Answers

Task 1: Article Summary

(a) Citation Details

- Authors: J. Yu, H. Li, S. Wang, et al.
- Title: *Deep Reinforcement Learning in Healthcare: A Review of Adaptive Treatment Strategies,*

  Clinical Decision Support, and Personalized Medicine
- Conference/Journal: IEEE International Conference on Bioinformatics and Biomedicine (BIBM),

  2021
- DOI: 10.1109/BIBM52615.2021.9669601 (doi.org in Bing)
- Domain/Application: Healthcare, clinical decision support, personalized treatment.
- ML Problem Addressed: Application of reinforcement learning (RL) to optimize treatment strategies and patient outcomes.

(b) Objective & Research Gap (150–200 words)

The article reviews how deep reinforcement learning (DRL) can be applied to healthcare for adaptive treatment strategies, clinical decision support, and personalized medicine. The authors identify a gap: traditional machine learning models in healthcare often rely on static datasets and cannot adapt to dynamic patient state transitions. RL, however, can model sequential decision-making, making it suitable for treatment planning where patient conditions evolve over time. The paper surveys recent RL applications in oncology, diabetes management, and critical care, highlighting challenges such as sparse rewards, limited patient data, and ethical concerns. The authors propose integrating electronic health records (EHRs) with RL frameworks to enable adaptive policies that improve patient outcomes. They emphasize the need for explainable RL models to gain clinician trust and suggest hybrid approaches combining RL with statistical learning for better interpretability.

Task 2: Methodology Analysis

(a) Techniques & Dataset

- ML Technique: Deep Reinforcement Learning (variants of Q-Learning, Deep Q-Networks, Policy Gradient methods).
- Dataset: Electronic Health Records (EHRs) from multiple clinical domains (ICU patient records, oncology treatment logs).
- Preparation: Data cleaning, normalization of lab values, encoding categorical features (e.g., treatment types), handling missing values.

(b) Mapping to CO4 Topics

- Topic: Reinforcement Learning (RL).
- Strength: Ability to model sequential patient state transitions and optimize long-term outcomes.
- Limitation: Requires large, high-quality longitudinal datasets; interpretability remains a challenge.

Task 3: Results & Critical Evaluation

(a) Reported Results

Model	Accuracy	AUC-ROC	Reward Convergence
Deep Q-Network (DQN)	~82%	0.87	Stable after 200 episodes
Policy Gradient RL	~80%	0.85	Slower convergence
Baseline Logistic Regression	~78%	0.82	Not applicable

(b) Critical Evaluation

- Realism: Metrics are promising but may be optimistic due to limited datasets.
- Bias Risks: Patient data may not represent diverse populations; evaluation protocols often lack external validation.
- Suggested Experiments: Cross-hospital validation, inclusion of multi-ethnic datasets, comparison with clinician decisions.

(c) Real-World Applicability

- Deployment: ICU treatment planning, oncology drug scheduling, diabetes management.

Challenges:

1. Data availability (EHR access, privacy).
2. Compute requirements for DRL training.
3. Ethical issues: patient safety, bias, explainability.

Task 4: Reflection & Learning Outcome

(a) Key Insights

- Sequential Decision-Making (RL) → Reinforcement learning uniquely models patient health transitions, deepening understanding of MDPs in clinical contexts.
- Explainability Challenge → Reinforces importance of interpretable ML, connecting to course topics on decision trees/statistical learning.
- Hybrid Approaches → Combining RL with logistic regression improves trustworthiness, linking to inductive/statistical learning concepts.

(b) Proposed Extension

- Novel Experiment: Apply DRL to chronic disease management (e.g., diabetes) using wearable sensor data.
- Justification: Continuous glucose monitoring provides rich sequential data, ideal for RL.
- Expected Impact: More personalized, adaptive treatment recommendations; improved patient adherence and outcomes.